Solving and Graphing One-Step Inequalities
Answer Key

Grade 6–7 Expressions and Equations

Quick Answers

1. $-\infty$, 5, ¡built-in function open¿
2. 8, ∞, *closed*
3. $-\infty$, -5, *closed*
4. 5, ∞, ¡built-in function open¿
5. $-\infty$, 6, *closed*
6. $-\infty$, -4, ¡built-in function open¿
7. -5, ∞, ¡built-in function open¿
8. $-\infty$, -3, *closed*
9. -6, ∞, ¡built-in function open¿
10. 7, ∞, *closed*, 7
11. $-\infty$, 9, *closed*
12. $-\infty$, -4, *closed*

Curriculum

Level: Grade 6–7 Expressions and Equations
7.EE.B.4 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 14 tagged checks.

1. Solve and graph $x + 4 < 9$.

Step 1. The variable has 4 added to it, so undo the addition: subtract 4 from both sides. Subtraction never moves the inequality sign.
Step 2. $x + 4 - 4 < 9 - 4$, which leaves $x < 5$.
Step 3. Graph: the sign is strict, so an *open* circle at 5, shaded to the left. Solution: $\boxed{x < 5}$
Check: $x = 4$ gives $8 < 9$ (true); $x = 5$ gives $9 < 9$ (false), so 5 itself is correctly excluded.

2. Solve and graph $n - 6 \geq 2$.

Step 1. Six is subtracted from n, so add 6 to both sides.
Step 2. $n - 6 + 6 \geq 2 + 6$, which leaves $n \geq 8$.
Step 3. Graph: $\geq$ includes the boundary, so a *solid* dot at 8, shaded to the right. Solution: $\boxed{n \geq 8}$
Common wrong answer: $n \geq -4$ — subtracting 6 instead of adding it. Undo the operation you see, do not repeat it.

3. Solve and graph $x + 7 \leq 2$.

Step 1. Subtract 7 from both sides to isolate x.
Step 2. $x \leq 2 - 7$, and $2 - 7 = -5$, so $x \leq -5$.
Step 3. Graph: solid dot at -5, shaded to the left. Solution: $\boxed{x \leq -5}$
Listen for: a negative boundary is not a reason to flip anything. The sign only moves when you multiply or divide by a negative.

4. Solve and graph $4x > 20$.
Step 1. The variable is multiplied by 4, so divide both sides by 4. Four is positive, so the sign stays as it is.
Step 2. $\frac{4x}{4} > \frac{20}{4}$, which leaves $x > 5$.
Step 3. Graph: open circle at 5, shaded to the right. Solution: $\boxed{x > 5}$

5. Solve and graph $\frac{m}{3} \leq 2$.
Step 1. The variable is divided by 3, so multiply both sides by 3 — again a positive number, so no flip.
Step 2. $3 \cdot \frac{m}{3} \leq 3 \cdot 2$, which leaves $m \leq 6$.
Step 3. Graph: solid dot at 6, shaded to the left. Solution: $\boxed{m \leq 6}$
Common wrong answer: $m \leq \frac{2}{3}$ — dividing by 3 again instead of undoing the division.

6. Solve and graph $-2x > 8$.
Step 1. The coefficient is -2, so isolating x means dividing both sides by a negative number — and that reverses the inequality sign.
Step 2. $\frac{-2x}{-2}$ and $\frac{8}{-2}$ give x and -4, and the $>$ becomes $<$: $x < -4$.
Step 3. Graph: open circle at -4, shaded to the left. Solution: $\boxed{x < -4}$
Common wrong answer: $x > -4$ — the sign was not flipped. Test $x = 0$: $-2(0) = 0$, and $0 > 8$ is false, so 0 must *not* be in the solution — which rules the un-flipped answer out immediately.

7. Solve and graph $y + 9 > 4$.
Step 1. Subtract 9 from both sides.
Step 2. $y > 4 - 9$, and $4 - 9 = -5$, so $y > -5$.
Step 3. Graph: open circle at -5, shaded to the right. Solution: $\boxed{y > -5}$

8. Solve and graph $6w \leq -18$.
Step 1. The coefficient is 6, a positive number, so divide both sides by 6 and leave the sign alone. The -18 on the right is not what decides this.
Step 2. $w \leq \frac{-18}{6} = -3$.
Step 3. Graph: solid dot at -3, shaded to the left. Solution: $\boxed{w \leq -3}$
Common wrong answer: $w \geq -3$ — flipped because a negative number appeared somewhere. The rule keys on what you divide *by*, not on the sign of the answer.

9. Solve and graph $-\frac{x}{3} < 2$.
Step 1. The left side is x divided by -3, so undo it by multiplying both sides by -3 — a negative multiplier, so the sign reverses.
Step 2. x on the left, $2 \cdot (-3) = -6$ on the right, and $<$ becomes $>$: $x > -6$.
Step 3. Graph: open circle at -6, shaded to the right. Solution: $\boxed{x > -6}$
Check: $x = 0$ gives $0 < 2$ (true), and 0 does lie to the right of -6, so the flipped direction is the right one.

10. Chess club: at least 25 members.
Step 1. "At least 25" means the total must be 25 or more, so the relation is $\geq$. The total is the 18 current members plus the m new ones: $m + 18 \geq 25$.
Step 2. Subtract 18 from both sides: $m \geq 25 - 18$, so $m \geq 7$.
Step 3. Graph: solid dot at 7, shaded to the right. Solution set: $\boxed{m \geq 7}$
Step 4. Part (b) asks for the smallest value that works. The boundary is included, and a club signs up whole people, so the answer is $\boxed{m = 7}$
Listen for: "more than 25" — "at least" includes 25, which is exactly why the dot is solid.

11. Concert tickets: at most 72 dollars.
Step 1. "At most" means the cost may reach the budget but not pass it, so the relation is $\leq$. The cost of t tickets is $8t$: $8t \leq 72$.
Step 2. Divide both sides by 8 (positive, so no flip): $t \leq \dfrac{72}{8} = 9$.
Step 3. Graph: solid dot at 9, shaded to the left. Solution: $\boxed{t \leq 9}$
Note for the reader: in context t also has to be a whole number that is 0 or more, so the practical answer is 0 through 9 tickets. The inequality itself is what the problem asked for.

12. Jordan's error on $-5x \geq 20$. *Part of this problem is an open explanation, flagged for manual review; the model wording below is what a full-credit sentence contains.*
Step 1. Divide both sides by -5. Because -5 is negative, the inequality sign must reverse at the same moment.
Step 2. $\dfrac{-5x}{-5} = x$ and $\dfrac{20}{-5} = -4$, and $\geq$ becomes $\leq$: $x \leq -4$.
Step 3. Graph: solid dot at -4, shaded to the *left* — the opposite ray from the one Jordan drew. Correct solution: $\boxed{x \leq -4}$
Step 4 (the explanation). Jordan divided by a negative number and kept the sign. Multiplying or dividing by a negative reflects every number to the other side of 0, so the number that used to be larger is now smaller and the order of the two sides is reversed. Concretely: $3 > 1$, but $-3 < -1$.
Full credit: the correct solution $x \leq -4$, a correctly shaded graph, *and* a sentence that says dividing by a negative reverses order. "You forgot to flip" names the rule but not the reason — push for the second half.